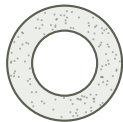


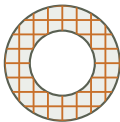
a What each acquisition supplies

Routine cine



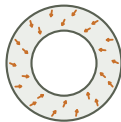
boundaries +
weak texture

Tagging



prepared grid
(material pattern)

DENSE

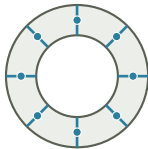


phase-encoded
displacement

Only tagging and DENSE carry a material label;
cine supplies boundaries and weak texture.

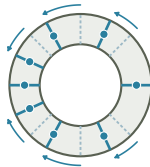
b Two mappings, identical contours

Mapping 1: $\theta' = \theta$



Dice = 1, Hausdorff = 0; $r' = r$, $a = 0.5$

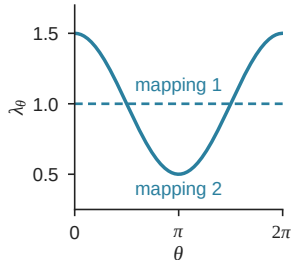
Mapping 2: $\theta' = \theta + a \sin \theta$



identical
masks

Dashed: reference positions. Teal: mapped material
points (same eight points in both panels).

c Resulting stretch



$\lambda_\theta = 1 + a \cos \theta$, range 0.5–1.5;
orientation preserved for $|a| < 1$

Analytic example, not a measured cardiac result. The stretch ratio λ_θ is not Green–Lagrange strain.